

2023



Guardant Health’s Data Platform on AWS Helps It Conquer Cancer with Generative AI

Learn how precision oncology company Guardant Health built a data platform powered by data mesh architecture on AWS.

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50,000

hours saved in manual labor (estimated)

Improved

disseminates information to physicians



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Protects

patient data with encryption



Accelerated

turnaround time to improve patient service

Overview

[Guardant Health](#) creates advanced, proprietary genomic sequencing tests that are designed to help every person live a life free from cancer. Data is at the core of the company's operations, with each of its 14 functional groups generating and analyzing a wealth of information.

To continue transforming lives, Guardant Health needed to improve data access and usage by converging data into meaningful, readily accessible knowledge. Using Amazon Web Services (AWS), it built a secure, unified data mesh system that decentralizes and automates data collection and analytics for its blood tests and other operational data. This self-service system helps different teams perform critical activities faster and contributes to better patient outcomes by helping physicians make informed decisions.



Opportunity | Turning to AWS to Create a Decentralized Data Solution for Guardant Health

Founded in 2012, Guardant Health uses proprietary tests, advanced analytics, and real-world data to support cancer patients and the physicians who treat them. The company's blood and tissue tests help improve patient outcomes across all stages of cancer, from screening for early detection and monitoring for recurrence to helping physicians select the best treatment for advanced cancer patients.



Guardant Health recognized that its data silos were stifling innovation. Despite unifying its data, inconsistent metric calculations across business groups led to varied results. The data platform, managed through a platform-as-a-service offering, could not scale effectively as the company's data grew by five times within 1 year. A more robust solution was necessary. "We wanted to build something fast with our limited staff resources," says Rajesh Kucharlapati, technology leader at Guardant Health. "The platform-as-a-service offering worked great for the first 6 months, but it was not built for sharing and scaling at the level at which we were growing."

To create a decentralized data-excellence environment, Guardant Health explored the concept of a data mesh architecture on AWS. This self-service approach encourages each team to focus on its specific area of expertise rather than extend themselves across multiple projects. "With data at the forefront on AWS, we can move faster, empower teams to tap into their unique expertise, and save time through automation," says Kucharlapati.

Solution | Using over 25 AWS Services to Build a Powerful System for Data and Analytics

Guardant Health worked closely with the AWS team to design the data mesh architecture and chose more than 25 AWS services to form the foundation of its data platform. This includes [AWS Glue](#), a serverless data integration service, and [Amazon Athena](#), a serverless, interactive analytics service. The platform also uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage service offering industry-leading scalability, data availability, security, and performance, and [AWS Lake Formation](#), a solution that creates secure data lakes, making data available for wide-ranging analytics.

The data platform went live in January 2021, strategically launched in phases to make sure that the core services ran smoothly. "We saw massive improvements, particularly with data ingestion," says Kucharlapati. "Previously, it happened in batches. The new platform does this in near real time."

Guardant Health restructured the data platform team into smaller, domain-aligned subgroups for accessing different AWS services and working with domain-specific data. For example, within a single domain team, a visualization expert might use [Amazon QuickSight](#), a service that provides unified business intelligence, to analyze their data. A data engineer may prefer Amazon Athena tables or [Amazon Redshift](#) to use SQL to analyze structured and semi-structured data.



The data platform team uses several AWS services to run and maintain interoperable data products for all domain teams and provide automation, development tools, and domain-agnostic functionality. For example, [AWS Fargate](#), a serverless compute engine, hosts their continuous integration and delivery solution. To protect health information and maintain HIPAA compliance, the team uses infrastructure as code to model, provision, and manage AWS resources using in-house templates that are preconfigured with a range of AWS services. For example, the infrastructure-as-code template for Amazon S3 is preconfigured with [AWS Key Management Service](#) (AWS KMS), which creates, manages, and controls cryptographic keys. AWS Glue templates are configured with [Amazon CloudWatch](#), which collects and visualizes near-real-time logs, metrics, and event data.

For advanced analytics and machine learning, Guardant Health uses [Amazon SageMaker](#), a service that is used to build, train, and deploy machine learning models for any use case, and [Amazon SageMaker JumpStart](#), a service that provides pretrained models. “With these cutting-edge capabilities, we are redefining the data landscape,” says Jaspinder Singh Virdee, solutions architect consultant at Guardant Health. “We are currently working on several use cases, including an image processing initiative to improve operational efficiency.”

Guardant Health built its real-world evidence product GuardantINFORM on top of the data platform to help its biopharmaceutical partners accelerate advanced cancer therapy development. “With better data discovery, thanks to an improved query engine and robust tools, we are empowering those who use our platform for research and development or clinical trials,” says Gautam Nayak, senior engineering manager at Guardant Health. “The datasets from GuardantINFORM offer recommendations on cancer therapy usage in the clinic, providing insights into tumor evolution and treatment resistance throughout each patient’s journey.

Outcome | Driving Improved Patient Outcomes with Automation, Generative AI, and Accelerated Decision-Making

With its data platform, Guardant Health estimates that it has saved 50,000 hours of manual labor across functional groups in 1 year. “The tools and capabilities that we have built on AWS empower our teams to accelerate the development of data assets and AI capabilities, resulting in faster insights and decision-making,” says Kucharlapati. “We are also optimizing our internal business processes by integrating AI solutions to enhance the likelihood of success within a shorter timeframe and at reduced cost.”

The robust data foundation is helping the company harness generative AI. Guardant Health is using [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing foundation models, to construct models grounded in genomic data. It is incorporating these innovations into patient outreach programs to expedite operations, achieve efficiency gains through automation, and empower domain teams to use their unique expertise.

By adopting a domain-focused approach, Guardant Health has scaled its data organization to over 70 members. As the system becomes more mature, more teams will start using it for external data collaborations and to become more data-centric—all to help cancer patients live longer and healthier lives.

About Guardant Health

Guardant Health is a precision oncology company that aims to help cancer patients live longer and healthier lives. It uses proprietary tests, advanced analytics, and digital solutions to support patients and physi



AWS Services Used

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models from leading AI companies via a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI

[Learn more »](#)

AWS Glue

AWS Glue is a serverless data integration service that makes it easier to discover, prepare, move, and integrate data from multiple sources for analytics, machine learning (ML), and application development.

[Learn more »](#)

AWS Lake Formation

AWS Lake Formation centralizes permissions management of your data and makes it easier to share across your organization and externally.

[Learn more »](#)

Amazon SageMaker JumpStart

Amazon SageMaker JumpStart provides pretrained, open-source models for a wide range of problem types to help you get started with machine learning.

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